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Patient-Reported Outcomes

Cross-Cultural Adaptation of the Beta EQ-5D-Y-5L Into Chichewa (Malawi)

Lucky G. Ngwira, MPH, Jennifer Jelsma, PhD, Hendramoorthy Maheswaran, MBBS, PhD, Fanny Kapakasa, BEd, Sarah Derrett, PhD, Stavros Petrou, PhD, Louis Niessen, PhD, Sarah C. Smith, PhD



ABSTRACT

Objectives: The EuroQol Group is developing a new EQ-5D-Y-5L version with 5 severity levels for each of the 5 dimensions. The 5 severity levels describe different health severities and there is a potential for severity level inversion. This article aims to report the process of cross-cultural adaptation of the beta EQ-5D-Y-5L into Chichewa (Malawi) using the card ranking exercise, which has been added to the EQ-5D-Y-5L translation protocol.

Methods: To assess the correct hierarchical ordering of severity levels, the adaptation followed the EQ-5D-Y-5L translation protocol. Cognitive interviews were undertaken to establish conceptual equivalence. Thereafter, 4 iterations of ranking exercises were conducted, leading to amendments of the translated Chichewa version to arrive at a final version.

Results: The iterations were assessed by 18 participants aged 8 to 14 years. Health proved to be a difficult concept to translate as was “discomfort.” Cognitive interviews identified further conceptual issues, particularly with the “looking after myself” dimension. Considerations about lack of soap or water indicated that some children did not fully comprehend this dimension as being about the ability to wash and dress themselves. The iterative card ranking exercise detected severity level inversion between “a little bit” and “some,” and between “a lot” and “extreme” and alternative Chichewa words/phrases were then tested. Ultimately, the intended hierarchical severity ranking was achieved and an acceptable Chichewa version was produced.

Conclusions: Conceptual and linguistic equivalence to the English EQ-5D-Y-5L was established for the Chichewa EQ-5D-Y-5L version. The card ranking exercise was instrumental in correcting severity level inversion and supporting the comprehensible translation.

Keywords: adaptation, childhood, EQ-5D-Y, EQ-5D-Y-5L, “health-related quality of life,” ranking exercise, “sub-Saharan Africa,” translation.

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Introduction

In 2010, the EuroQol Group developed a child-friendly EQ-5D-Y to measure health-related quality of life (HRQoL) in children and adolescents across 5 dimensions (“mobility,” “looking after myself,” “doing usual activities,” “having pain or discomfort,” and “feeling worried, sad or unhappy”), with 3 severity levels (eg, no problems, some, a lot of problems) indicating the degree of problems experienced in each of the dimensions.¹ The EQ-5D-Y has, however, been criticized for only containing 3 severity levels, and for being insensitive to small changes in health status and exhibiting ceiling effects.²

To overcome this, the EuroQol Group simultaneously developed in 4 languages (German, Spanish, Swedish, and English) a beta (trial) version—the EQ-5D-Y-5L.³ The EQ-5D-Y-5L is an extension of the EQ-5D-Y with 5 severity levels (eg, no problems [or not], a little, some [or quite], a lot [or really], and cannot do [or extreme or extremely]) in each dimension. An experimental EQ-5D-Y-5L version has also been evaluated in Chinese for Hong

Kong using a standard EuroQol translation protocol.⁴ The adult EQ-5D-3L has become one of the most widely used HRQoL measures.⁵ The EQ-5D-Y was developed in 2010, and although it has not been used as widely as the adult EQ-5D-3L, it has the potential for increased usage.⁶ The EQ-5D-Y-5L has been developed to have more response options to increase sensitivity and resolve ceiling effects observed with the EQ-5D-Y.^{2,3} An increased number of severity levels means the EQ-5D-Y-5L can more comprehensively describe HRQoL in children and adolescents. It is, therefore, imperative to establish whether the beta EQ-5D-Y-5L achieves conceptual equivalence in other settings and languages, including those of sub-Saharan Africa.

To ensure that the beta (UK English) EQ-5D-Y-5L is appropriate for use in other settings, the translated versions must be linguistically and conceptually equivalent,⁷ and there should be no issues of severity level inversion.⁸ Severity level inversion is when a worse severity level is ranked higher than a better severity level, for example, “a lot of problems” is preferred to “having a little bit of problems.” A key requirement for HRQoL measures is that the

severity levels should be discrete, and participants should be able to rank them hierarchically in order of severity.⁸ If this is not resolved, it could eventually lead to preference inversion in self-reported health and a utility valuation exercise.⁹

Preference-based HRQoL measures play a vital role in economic evaluation and policy decision making. The Malawi health sector uses the National Health Policy (2018-2030) as a long-term strategic document and the Health Sector Strategic Plan II (2017-2022) as a medium-term strategy. The goal is to achieve universal health coverage and health financing plays a critical role in the achievement of universal health coverage goals and objectives. An economic evaluation will, therefore, become important in determining programs for health financing and application of the adult EQ-5D-3L (and EQ-5D-5L), EQ-5D-Y, and EQ-5D-Y-5L could be key instruments for assessing whether these goals are achieved. Malawi's population is increasing, about half being below the age of 15 years, and at the current growth rate, it is projected to double by 2038.¹⁰ The focus on attaining Sustainable Development Goals 3 means that health programs will not only aim to reduce childhood mortality but also improve children's HRQoL, and this requires robust instruments.¹¹ Currently, only the adult EQ-5D-3L is available in the Chichewa language for Malawi. Chichewa is the national language of Malawi (with a population of 18 million),¹² and is also spoken in the neighboring countries of Mozambique and Zambia.¹³

The standard EQ-5D translation protocol includes translation (forward and backward), and cognitive interviews to establish linguistic and conceptual equivalence.¹⁴ As the EQ-5D-Y has 3 severity levels, this protocol does not involve a specific detailed assessment of severity level inversion. In view of this, the EuroQol's Version Management Committee (VMC), the body within EuroQol that is responsible for overseeing translations of all instruments developed by the EuroQol, has developed a card ranking exercise for the beta EQ-5D-Y-5L. Previously, a rating by children was found useful in the development of the beta EQ-5D-Y-5L severity level wording.³ A card ranking exercise developed by the VMC, similar in principle to the rating tasks, has been added to the standard EQ-5D-Y cross-cultural translation protocol for the beta EQ-5D-Y-5L (hereby referred to as draft EQ-5D-Y-5L translation protocol).¹⁵ The card ranking exercise has been developed to assess the suitability of the translated wording of the 5 severity levels for children. The ranking exercise allows assessment of severity level inversion during the translation process by asking children to rank EQ-5D-Y-5L severity levels from least to most severe problems.

Aim

This study aimed to cross-culturally adapt the EQ-5D-Y-5L for use in Malawi using the draft EQ-5D-Y-5L translation protocol.

Methods

Participants and Recruitment

This study with healthy and sick children was undertaken in Blantyre, Malawi between March 2019 and December 2019. Blantyre is the commercial capital of Malawi and comprises an urban population. The healthy children consisted of a convenience sample of children attending primary schools within Blantyre. The sick children consisted of a convenience sample of children visiting the outpatient department at the Queen Elizabeth Central Hospital in Blantyre. Queen Elizabeth Central Hospital is the largest referral and teaching hospital in Malawi.

Written assent and consent were obtained from both children and parents/guardians, respectively, for each interview. For sick participants, the invitation came at the end of clinical care. For healthy participants, invitations were through the school with a teacher. For both sets of participants, once consent was obtained, the questionnaires were distributed by the research team at the end of clinical care or on an arranged school day for interviews.

Instruments

The EQ-5D-Y-5L consists of a descriptive system with the same 5 dimensions as the EQ-5D-Y: "mobility," "looking after myself," "doing usual activities," "having pain or discomfort," and "feeling worried, sad or unhappy" (and EQ-5D-Y-5L visual analog scale [EQ VAS]), each with 5 severity levels: for example, (1) "no problems/not," (2) "a little bit of a problem," (3) "some problems/quite," (4) "a lot of problems/really," and (5) "extreme problems/extremely/cannot." Self-rated health is also assessed with an EQ-VAS on a "thermometer-like" vertical scale with scores ranging between 0 (worst imaginable health) and 100 (best imaginable health). The adult EQ-5D-3L Chichewa version has been validated for orthopedic patients in Malawi.¹⁶ The adult EQ-5D-3L Chichewa version and the English-language UK EQ-5D-Y source version were provided by the EuroQol to be used in the translation. Apart from differences in age-appropriate language, the main difference between all these instruments is that beta EQ-5D-Y-5L has 5 levels of severity whereas the others have only 3. The translators strove for consistency between the EQ-5D-Y-5L and the other versions as far as this was appropriate.

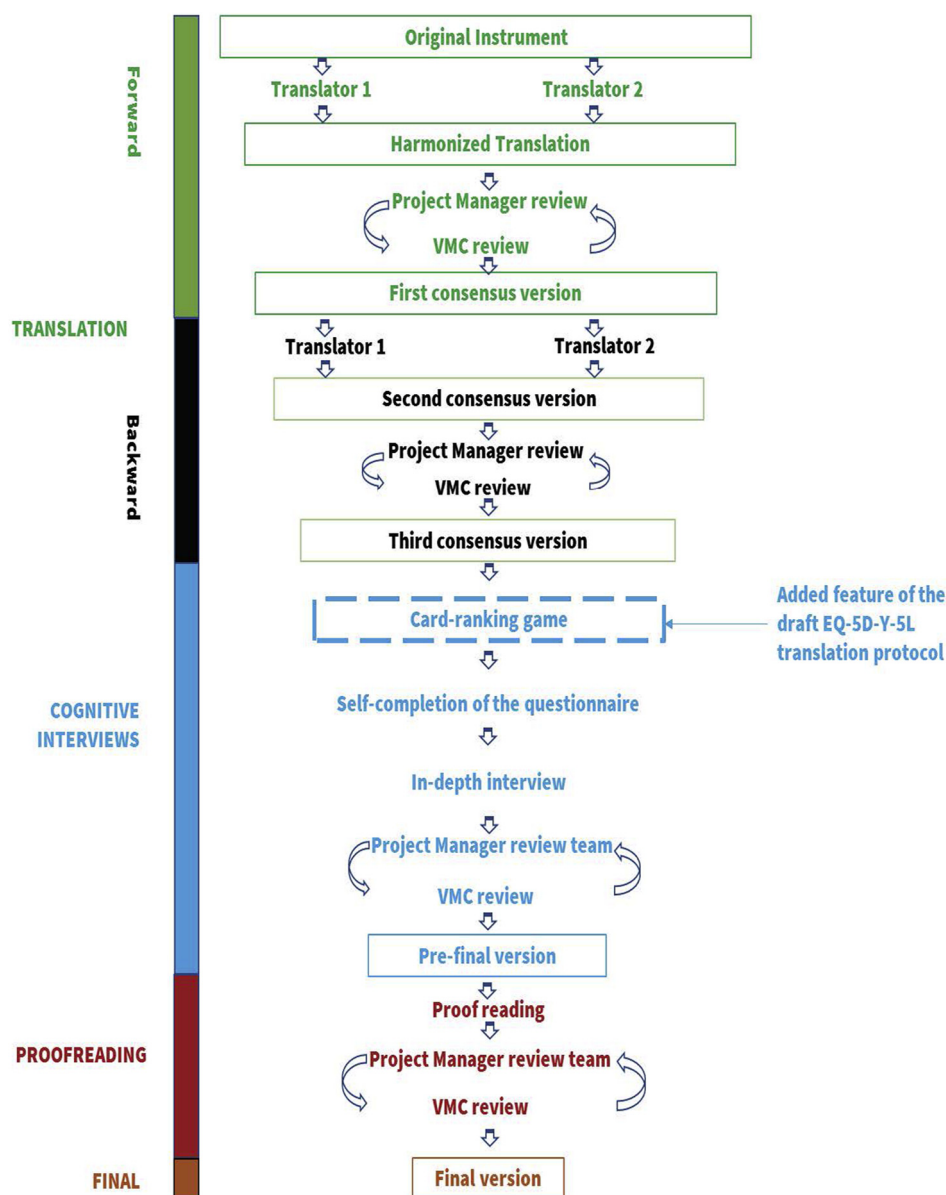
Procedure

Translation process

We undertook the translation process of the EQ-5D-Y-5L into Chichewa using the draft EQ-5D-Y-5L translation protocol including the card ranking exercise (Fig. 1). Forward translations from English to Chichewa were undertaken by experienced local translators, whereas translations from Chichewa back to English were undertaken by international volunteers (whose first language is English but also conversant with Chichewa). First, 2 sets of translators independently forward and backward translated the EQ-5D-Y from English to Chichewa and back to English. The forward and back translation of the EQ-5D-Y-5L included additional text in the EQ-5D-Y-5L. The EQ-5D-Y-5L severity level descriptions were discussed by translators and the lead researcher to reach consensus versions. The versions were forwarded and discussed iteratively with the VMC reviewer leading to ultimate approval. The approved translated questionnaire was then piloted through cognitive interviews.

Cognitive interview process

The cognitive interviews were conducted to establish the conceptual equivalence of the concepts in the translated Chichewa and English languages. The cognitive interviews, which included the ranking exercise, were conducted in 4 rounds made up of different participants each time. The first round of the interviews was conducted with 4 healthy and 3 sick children aged 8 to 15 years according to the VMC translation protocol. Nevertheless, to resolve any apparent severity level inversion, a further 3 subsequent cognitive interviews were conducted with healthy children alone; it was not considered likely that severity inversion would relate to the health status of participants. The participants took part in the following 3 cognitive interview steps: (1) card ranking exercise (a new translation protocol feature) to evaluate the comprehension of the EQ-5D-Y-5L questionnaire; (2) self-

Figure 1. EQ-5D-Y-5L cross-cultural translation process.

EQ-5D-Y-5L indicates 5 level EQ-5D-Y; VMC, Version Management Committee.

completion of the EQ-5D-Y-5L questionnaire, and (3) in-depth interviews. The first 10 participants took part in all the 3 steps of cognitive interviews. The rest ($n = 8$) of the participants only took part in the card ranking exercise because it was seen that no new information was emerging from the last 2 completed cognitive interviews.

Card ranking exercise. The card ranking exercise was specifically developed by the VMC to detect severity level inversion in the beta EQ-5D-Y-5L so that this could be investigated and corrected during the adaptation process.¹⁵ The card ranking exercise involved children ranking sets of 5 cards describing the translated severity levels from least to most severe problems. The 5 ranked card sets consisted of an introductory (warm-up) set, and then 4 of the 5 EQ-5D-Y-5L dimensions. The EQ-5D-Y-5L translation protocol has, by design, 4 of the 5 dimensions.¹⁵ This is because all 5 severity levels have the same qualifiers for

the “mobility,” “looking after myself,” and “doing usual activities” dimensions. In contrast, the “having pain or discomfort” dimension has the same qualifiers as the first 3 dimensions, except for the fifth severity level. The “feeling worried, sad, or unhappy” dimension also has unique qualifiers for the third to fifth severity levels. Owing to the similarity of qualifiers for all the severity levels for the first 3 dimensions (mobility, doing usual activities, and looking after myself), the VMC decided not to keep the “doing usual activities” dimension as part of the card ranking exercise.¹⁵ At the start of the interviews, participants first ranked the introductory set and any issues with understanding the card ranking process were clarified before moving on to the other 4 sets. Interviewers, with previous interview guides, were interested in observing whether or not the intended card order of the translated Chichewa cards was in the order anticipated (from no problems to extreme problems).

Self-completion of the EQ-5D-Y-5L questionnaire. After the card ranking exercise, the participants completed the Chichewa EQ-5D-Y-5L questionnaire. By design, the Chichewa EQ-5D-Y-5L was administered to the children after the card ranking exercise. This was done to avoid any possible biases that may have emerged from children seeing the intended ordering of the severity levels when self-completing the EQ-5D-Y-5L. If necessary, children who struggled to understand the instructions for self-completion of the questionnaire were given help to understand the task. Nevertheless, the children were not given help to respond to the questionnaire. Any difficulties that the children experienced when self-completing the questionnaire were noted by the interviewer. Additionally, any assistance given to help them understand the instructions was also recorded on a separate sheet of the article.

In-depth interviews about the EQ-5D-Y-5L questionnaire. Each child participating in the first 2 rounds of interviews then took part in a one-to-one interview about his/her understanding of the questionnaire. The interview questions were open-ended and were adapted from the draft EQ-5D-Y-5L translation protocol.¹⁴ Participants were first asked to comment on: (1) the comprehensibility of the whole questionnaire, (2) words they found difficult to understand (in the dimensions, severity levels, and instructions), and (3) what they would change/add to make the instructions easy to understand. Additionally, the participants were asked to reflect on how they understood each question and severity level, and subsequently provide real or hypothetical examples of people experiencing specific severity levels. Other than comprehension of instructions and wording of the questionnaire, participants were not asked regarding the survey in general.

At the end of the interviews, the children were thanked for their participation and were compensated with an exercise book and a pen/pencil for their time.

Data Analysis

For the card ranking exercise, individual severity level ordering by each child and any observed difficulties to do so were recorded on a data sheet by the interviewer, which was later summarized into a chart for all participants. The correct rank orderings were given an orange color whereas a blue color represented incorrect rankings. The blue squares indicated where the problems were that needed addressing for the next iteration. For the self-completion exercise and in-depth interviews, the recorded text was similarly put in a table format with a comments section. The comments section provided guidance on the part of the text that needed changing for the next iteration.

Subsequently, the translated wordings were revised following each iteration of cognitive interviews until all inconsistencies were resolved. This version was then proofread for typographical and layout errors. The prefinal version was forwarded to the VMC and EuroQol Office for endorsement as the final translated Chichewa (Malawi) beta EQ-5D-Y-5L version.

Results

Participants and Recruitment

The cognitive interviews were carried out in 4 separate rounds. The first round involved 7 children (4 healthy and 3 sick) as per the EQ-5D-Y-5L translation protocol. Each of the next 3 rounds comprised 4 (all healthy) children. All participants were attending primary school, and none were already familiar with the EQ-5D-Y-

Table 1. Demographic characteristics of the interviewed participants.

Characteristics	n
Participants*	18
Gender	8 girls, 10 boys
Age in years	
8-10	13
11-14	5
Health condition	
Acute	3
Healthy	15

*n = 1 has been excluded.

5L. In total, 19 children (10 boys, 9 girls) aged 8 to 14 years (mean 9.6 years), were interviewed over the 4 rounds (Table 1). After consultation with the VMC reviewer, data from the first respondent was removed from the results because of some confusion the child had in understanding the task. This confusion could have arisen from any of the following: (1) this was the first interviewee (and it is possible that the interviewer did not clearly articulate instructions); (2) the child may have been confused by the task; or (3) a combination of both. Results from the remaining 18 children (3 sick [all acutely ill] and 15 healthy) are presented.

Translation

There were several conceptual and linguistic problems identified from the translation process. Most of these were identified and resolved by translators during the first consensus process. Nevertheless, translation issues, particularly on qualifiers for the severity levels, were resolved during the ranking exercise and cognitive interviews. The results of the cognitive interviews are discussed below. Regarding translation, the first amendment was from the Chichewa adult EQ-5D-3L version. The bracketed (Nyanja) was removed from the heading Chichewa (Nyanja) for Malawi' as Nyanja is longer spoken in Malawi but some neighboring parts of Zambia and Mozambique. This may not have altered the concept but was done to be consistent with the language being translated into.

Conceptual issues

Health. From the outset, it became apparent that the translation of the term "health" was a problem. Health was translated differently by translators as "umoyo," which is general health; "moyo," which is life; and "thanzi," which is about being healthy. This was discussed at length in the consensus discussion with the translators, and the agreement was that "moyo" is about being alive and not necessarily about being healthy (a good life) or health per se (umoyo). "Umoyo" was agreed upon as the correct translation for the term "health" through group consensus among the translators.

Pain or discomfort. The adult Chichewa EQ-5D-3L version had translated "pain" as "ululu." This word does mean pain, but implies extreme pain, and is not frequently used by children. Thus pain was translated as "kuphwanya m'thupi" (body pain) in the EQ-5D-Y-5L. The aspect of physical pain was added because pain can also be interpreted as mental anguish and the VMC's definition of terms indicates that pain is intended to refer to physical pain. Additionally, the conjunction "kapena" ("or") seems not to have the same effect as in English, in which it separates "pain" from "discomfort." So "m'thupi" (body) was

added to the descriptor even though it appears repetitively in the same sentence. This is not problematic as this is often the case in the Chichewa language.

“Discomfort” was translated as “kusamva bwino” in contrast with “kuphwanya m’thupi kosowetsa mtendere” in the adult EQ-5D-3L version. We opted for the former because “kuphwanya m’thupi kosowetsa mtendere” conveys a pain of greater intensity, which differs from the VMC’s intended definition of discomfort as “uncomfortable physical sensation of a lower grade of intensity than pain.”¹⁴

Semantic issues

Other deviations from the existing Chichewa adult EQ-5D-3L version included changing the expression “I have (no problems)...” that was translated “*Ndimakhala ndi...*” in the adult version to “*Ndili ndi...*” This is because “*Ndimakhala ndi...*” carries past/present continuous action—that is, how the person is (always has been) up to this very moment. To a child, this does not reflect how one feels on a particular day (“today”). Other phrases explored and rejected included “*Ndili ndi vuto*,” which does imply a problem with performing a particular task because of (in)ability to do so but not because of one’s attitude (child being disobedient for example) or other issues. “*Ndikuvutikirapo*” (I have a problem) was rejected as it refers to a struggle. “*Ndili ndi...*” was, thus, retained as the most accurate translation.

Additionally, the severity qualifier “some problems” is translated to “*mavuto ena*” in the adult EQ-5D-3L version. Nevertheless, “*ena*” relates to “other,” and “*mavuto ena*” could imply other problems on top of, or in addition to, the current problem relating to walking. “*Ena*” was, therefore, removed in the severity level.

We also included the qualifier for each of the discrete concepts in the “having pain or discomfort,” and “feeling worried, sad, or unhappy” dimensions. For example, the English version reads “I am a bit worried, sad, or unhappy” and this is understood to mean a bit relating to “feeling worried, sad, or unhappy.” Nevertheless, this could be quite confusing in the Chichewa version as it could be read that “a bit” applies to “worry” only. To overcome this potential problem, we included the qualifier for each of these concepts so that the reader is clear that this applies to “feeling worried, sad, or unhappy.”

Cognitive Interviews

A summary of the cognitive interviews is provided in Table 2.

Card ranking exercise

In the first round, each of the 6 participants ranked 5 severity levels for the 4 dimensions to give a total of 120 rankings (Table 3). Of these, 43% (52 of 120) across all dimensions were incorrect rankings (from the intended order). Of the incorrect rankings, 62% (32 of 52) were attributed to severity level inversion between the Chichewa wording for “a little bit” and “some.” It was evident that “a little bit,” initially translated “*pang’ono kwambiri*,” was the source of the confusion. The Chichewa words “*pang’ono*” and “*kwambiri*” mean “less” and “a lot,” respectively. Whereas the combination of these 2 words was supposed to imply “a lot less,” it was interpreted as “a lot more,” and led participants to interpret severity level 2 as worse than that of level 3. A further 14% of incorrect rankings were because of severity level inversion between the Chichewa words for “a lot” (6%) and “extreme” (8%). The rest of the incorrect rankings were made up of a mixture of incorrect ordering among all the severity levels. As there were several cases of severity level inversion between “a little bit” and “some,” this was revised and then retested.

Table 2. Cognitive interview summary.

Characteristics	Values	n
No. of interviews conducted, n*	19	18
No. of interviews that included all 3 activities ([1] ranking game; [2] self-completion of questionnaire; and [3] in-depth interviews)	11	10
The average time taken to complete all 3 activities, minutes	31.25	
Needed help to self-complete questionnaire	7	7
No of interviews involving card ranking exercise only, n	8	8
The average time taken to complete the ranking exercise, minutes ¹	22	22

No. indicates number.

*Number of participants included in all 3 activities but excluded from further analysis equals 1.

¹Based only on the last 8 participants as the first 10 involved all 3 activities.

The retesting in round 2 indicated that the change in translation generally resolved the severity level inversion for levels 2 (a little bit) and 3 (some). The severity level inversion was only evident in one respondent for both “having pain or discomfort” and “feeling worried, sad or unhappy,” and given that this was equal to only 2 responses, it was judged that this change had satisfactorily resolved the problem. Nevertheless, half of the incorrect rankings in round 2 were because of severity level inversion between the qualifiers for levels 4 (a lot) and 5 (extreme), which reemerged in all dimensions except for the “looking after myself.” It was discovered that the translated qualifier for severity level 4 (*kwambiri*) was generally understood to mean “extreme” by children because in their everyday usage *kwambiri* is understood to mean extreme. The severity level 5 qualifier was, therefore, changed to “*moonjeza kwambiri*” to emphasize the intensity, and this was then retested during the third round of iterative interviews.

In round 3 of testing, 12 of all 28 incorrect rankings were still because of translated severity level inversions between qualifiers for levels 4 (a lot) and 5 (extreme). The latter was, therefore, further changed to “*kwambiri zedi*” for the “having pain or discomfort” and “feeling worried, sad, or unhappy” dimensions and then retested. The qualifiers for the other 3 dimensions were not changed because they have a different qualifier for level 5 (cannot), which was not problematic for the translated version.

The final round of testing showed that the changes made to the qualifier for extreme severity level 5 had resolved the severity inversion for levels 4 and 5. At this point, the EQ-5D-Y-5L translated version was deemed to have established appropriate hierarchical ordering. The translated questionnaire was then ready for proofreading ahead of self-completion and in-depth interviews.

Self-completion of the questionnaire

We observed that appropriate self-completion of the questionnaire without interviewer assistance was problematic among some children in this setting. A total of 7 of 10 children who were aged 8 to 10 years tended to tick every box instead of choosing 1 option for each of the 5 dimensions, which could indicate a lack of understanding of the instructions. Nevertheless, once the interviewer read and explained the instructions clearly to these children, they were able to appropriately choose 1 severity level per dimension (Table 4).

Table 3. Participant rankings of the card ranking exercise progressing over 4 rounds.

Beta EQ-5D-Y-5L dimensions*	Box number card is placed into																	
	Round 1						Round 2				Round 3				Round 4			
	R1	R2	R3	R4	R5	R6	R7	R8	R9	R10	R11	R12	R13	R14	R15	R16	R17	R18
Set 1 (pain or discomfort)																		
No pain or discomfort	1	1	1	1	1	1	1	1	1	1	1	3	2	1	1	1	1	1
A little bit of pain or discomfort	3	3	3	3	4	2	2	2	2	3	2	2	1	2	2	2	2	2
Some pain or discomfort	4	2	2	2	2	3	3	3	3	2	3	1	3	3	3	3	3	3
A lot of pain or discomfort	5	4	4	4	5	4	5	5	4	4	5	5	5	4	4	4	4	4
Extreme pain or discomfort	2	5	5	5	3	5	4	4	5	5	4	4	4	5	5	5	5	5
Set 2 (worried, sad, or unhappy)																		
Not worried, sad, or unhappy	1	1	1	1	1	1	1	1	1	1	1	3	1	1	1	1	1	1
A little bit worried, sad, or unhappy	3	3	3	3	4	3	2	3	2	2	2	2	3	2	2	2	2	2
Quite worried, sad, or unhappy	2	2	2	2	2	2	3	2	3	3	3	1	2	3	3	3	3	3
Really worried, sad, or unhappy	4	5	4	4	5	4	4	4	5	4	4	5	4	4	4	4	4	4
Extremely worried, sad, or unhappy	5	4	5	5	3	5	5	5	4	5	5	4	5	5	5	5	5	5
Set 3 (walking about)																		
No problems walking about	1	1	1	1	1	1	1	1	1	1	1	5	3	1	1	1	1	1
A little bit of a problem walking about	5	3	3	3	3	2	3	2	2	2	2	2	2	2	2	2	2	2
Some problems walking about	2	2	2	2	2	3	4	3	3	3	3	1	1	3	3	3	3	3
A lot of problems walking about	3	5	4	4	4	4	5	5	4	4	4	4	4	5	4	4	4	4
Cannot walk about	4	4	5	5	5	5	2	4	5	5	5	3	5	4	5	5	5	5
Set 4 (washing or dressing)																		
No problems washing or dressing himself/herself	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
A little bit of a problem washing or dressing himself/herself	2	3	3	3	2	3	2	2	2	2	2	5	2	2	3	2	2	2
Some problems washing or dressing himself/herself	3	2	2	2	3	2	3	3	3	3	3	3	3	3	2	3	3	3
A lot of problems washing or dressing himself/herself	4	4	4	4	4	4	4	4	4	4	5	2	4	4	4	4	4	4
Cannot wash or dress himself/herself	5	5	5	5	5	5	5	5	5	5	4	4	5	5	5	5	5	5

R indicates responder.

*R dimension responses reported in English rather than Chichewa for ease of comprehension; Blue coloring highlights the severity levels in which the child has ordered the cards in a manner not intended by the translators. Yellow coloring highlights correct card ordering.

Table 4 also shows that the children with health conditions reported suboptimal health states, and reported lower scores on the EQ-VAS scale compared with healthy children. In contrast, all healthy children reported perfect health on the descriptive system and corresponding high EQ-VAS scores (in which 1 child reported 95 and 6 children reported 100). One sick child with a poor health state of 32 132 had a high EQ-VAS score of 95,

whereas another sick child with a better health state of 11 222 provided a lower (worse) EQ-VAS score of 60.

In-depth interviews

Similar to the translation work, it emerged during cognitive interviews that some children struggled to distinguish health, (umoyo), healthy (thanzi), and life (moyo). Most participants

Table 4. Responses of the participants when completing EQ-5D-Y-5L.

Beta EQ-5D-Y-5L dimensions	Respondents*									
	R1	R2	R3	R4	R5	R6	R7	R8	R9	R10
Descriptive health profile summary [†]	11111	11111	11111	32132	11222	21111	11111	11111	11111	11111
EQ-VAS score	100	100	100	95	60	100	95	100	100	100

EQ-VAS indicates EQ-5D-Y-5L visual analog scale; R, responder.

*Whereas 18 participants completed the ranking exercise, only the first 10 were administered the EQ-5D-Y-5L.

[†]Where “11111” represents “No problems” reported for all 5 dimensions, and where “55555” would represent “Extreme problems” on all 5 dimensions.

mentioned that “umoyo” was about how one’s daily life was/how the body was feeling compared with “moyo,” which was the state of being alive, that is, opposite of death. Some of the participants were not able to draw this distinction between these 3 terms, and a few others expressed no opinion regarding this but agreed that this was about health. Therefore, the word “umoyo” was kept, as decided during the translation process.

Despite this, we observed that most participants were able to give appropriate examples of the different severity levels. For example, when it came to examples to illustrate “a little bit,” “some,” and “a lot” of problems in the “mobility” dimension, the children described such examples as someone with a sprained knee, an injured leg, and having mobility issues because of some health problems, respectively. For “a little bit” in the “doing usual activities” dimension, children gave examples of someone unwell but able to go to school; and similarly, appropriate examples were given for “a little bit,” “some,” and “a lot” of problems in the “having pain or discomfort” dimension. Examples such as someone with an upset tummy or with tonsillitis indicated appropriate comprehension for “having pain or discomfort.” It was also very clear that most children interviewed understood the “feeling worried, sad, or unhappy” concept as evidenced by examples given for this dimension. Participants cited that losing a friend or losing a loved one to death, lack of basic needs or money, lack of food at home, or home-based violence, or lack of school materials would result in different levels of worry, sadness, or unhappiness.

There were also some inaccurate responses. In the “looking after myself” dimension, 2 of the 10 participants gave examples of severity levels relating to lack of provisions such as soap or water as examples of “some problems.” In the “having pain or discomfort” dimension, 2 different participants gave examples that related to the loss of a loved one in death (emotional pain) or a friend that is sick and could not attend school as things that would make someone “having pain or discomfort.” Another child gave an example for “a lot” of “having pain or discomfort” of an adult diagnosed of a certain disease (eg, epilepsy) and, therefore, unable to do usual activities. From a child’s perspective, an epileptic person would be affected in physical functioning, but this example may pertain more to an emotional problem as opposed to bodily pain.

At the end of the interviews, when asked to make suggestions to change the wording of dimensions or severity levels, at no point during the in-depth interviews did the participants make additional meaningful suggestions to further improve the Chichewa wording of dimensions and severity levels.

Discussion

Ultimately, a version of the beta EQ-5D-Y-5L was developed in the Chichewa language. The version established acceptable linguistic and conceptual equivalence with the original United

Kingdom version. Nevertheless, there were a few issues that needed to be resolved during the adaptation process.

The concept of health was difficult to translate as 3 terms are used interchangeably. This is, however, problematic as these concepts are rather different. This confluence of health and life concepts is also common to other Bantu languages, possibly because of different views or models of health existing in these settings.¹⁷ In Zulu, “health” and “life” are both translated with “Mpilo” and in Kinyarwanda “ubuzima” can be used for both. One study in Kenya found that this was the case when they tried to disentangle health.¹⁸ So, we conclude that, whereas the children generally did seem to understand health, we recommend that the concepts of health and life in these Bantu cultures should be explored further, perhaps through focus group discussions to elicit more views from the target population.

The validity of including composite dimensions in the EQ-5D instruments has been questioned.^{19,20} As the items are entrenched in the measure (which has been universally applied²¹), however, no changes were made to the constructs included in the “having pain or discomfort” and “feeling worried, sad, or unhappy” dimensions. To highlight that all 2 or 3 components of these dimensions were highlighted and given equal prominence, we found it helpful to add the qualifier at each component to emphasize that it referred to every construct. For example, the severity level “I have a little bit of pain or discomfort” could imply that “a little bit” applies only to pain and not to both pain and discomfort. This was altered to read, “I have a little bit of pain or a little bit of discomfort.” This solution is not unique and has been incorporated into other translations, such as the Krio self-complete EQ-5D-3L and the Arabic for Jordan EQ-5D-Y versions (Personal communication, VMC, 2021). We recommend that future cross-cultural adaptation in other contexts similarly explore the need to emphasize all components in composite dimensions to maintain the correct meaning.

The Chichewa EQ-5D-Y-5L was found to be generally difficult to self-complete. We found that younger children, in particular, needed some interviewer assistance to accurately complete the questionnaire. For example, younger children tended to tick every severity level response option instead of choosing one, unless the interviewer assisted. Consequently, expanding the instructions could solve this problem. Users of the Chichewa EQ-5D-Y-5L might also consider using the interviewer-administered rather than the self-complete version for children younger than 12 years and particularly those younger than 10 years.

In addition, although we were generally satisfied that this Chichewa version established both linguistic and conceptual equivalence with the English UK version, caution in administering this version is advised. Whereas most participants gave specific examples that fittingly explained their understanding of the severity levels and dimensions, there were helpful lessons learned from in-depth interviews with a few participants who struggled to understand some dimensions. This also highlighted contextual

differences in terms of how people understand translated dimensions of HRQoL instruments. For the EQ-5D-Y-5L, we particularly observed difficulties concerning “looking after myself.” In high-income countries, washing is simply the action of washing the body. In low-income countries, the availability of water and soap may be considered by the participants rather than simply the ability to wash and dress themselves. This was also highlighted during the exercise in Sierra Leone to translate the EQ-5D-Y into Krio (Personal communication, VMC, 2021). The inclusion of needing to get water—that is, from a well or river—in the activity of self-care was highlighted in that translated version. It was recommended that the EuroQol Group should provide further guidance on environmental factors in the translation protocol. We believe the omission of such clarification as water availability and “mobility” issues might affect item equivalence, particularly in some low-income countries.²² Our results endorse that recommendation.

“Having pain or discomfort” has proven somewhat problematic in the past, firstly because these constructs, which are meant to refer to physical pain or discomfort, are often interpreted as emotional or mental pain or discomfort. Our results similarly showed that some children made reference to emotional or mental pain instead of physical. This finding was curious because we had included “m’thupi”(body) in the descriptor to establish conceptual equivalence with the concept of bodily pain. It is possible that the words used for discomfort (*kusamva bwino*) could perhaps have led others to conclude that discomfort was about emotional pain as opposed to uncomfortable physical sensation, which is the concept here. The descriptor for discomfort in the adult Chichewa EQ-5D version is “*kuphwanya m’thupi kosowetsa mtendere*,” which would seem to indicate pain of greater intensity as opposed to pain of lower intensity, which is the concept for discomfort here. We, therefore, changed the descriptor for discomfort in the youth Chichewa version to “*kusamva bwino*,” which means to feel uncomfortable. The “having pain or discomfort” dimension has proved difficult to translate in the past. In the Shona translation of the adult EQ-5D-3L, it was reported that such an expression as “I cannot even stay in one place” was considered for translation. In addition, discomfort was not regarded as a sensation separate from pain,²³ and, as such, the phrase “*kurwadziwa*” for pain was used. We would, therefore, recommend that future users of the youth Chichewa EQ-5D-Y-5L version should explain to participants (through interviewer assistance) that the discomfort mentioned here refers to physical sensation. Another option would be to include “physical” in the descriptor so that the dimension reads “physical pain or physical discomfort.”

The ranking exercise was instrumental in identifying translation problems in understanding the qualifiers for the 5 different severity levels. We believe that had it not been for the card ranking exercise, coming up with accurate qualifiers to establish linguistic equivalence and hierarchical ranking would not have been adequately established. As noted previously, severity level inversion in the questionnaire is a threat to the validity of valuation sets. If the severity level is either not properly understood or does not accurately distinguish between levels, it could lead to preference inversion in self-reported health and utility valuation exercises. This has been previously reported as a problem.⁹ Thus, we highly recommend the use of the ranking exercise when adapting the EQ-5D-Y-5L cross-culturally.

Finally, regarding possible differences between the EQ-VAS score and that associated with the descriptive system health state, we would not want to overinterpret this. There were only 2 out of 3 participants who had EQ-VAS scores that seemed inconsistent with their reported descriptive system health state. We would, however, recommend that future studies should explore which other aspects of “health” may be important to children beyond the 5 EQ-5D-Y-5L dimensions.

A limitation of our assessment of the Chichewa version of the EQ-5D-Y-5L was in the sample of participants. Collectively, there were more healthy than sick children involved in the other rounds of the cognitive interviews compared with the first one.¹⁴ In a South African study, it was found that the psychometric properties of the EQ-5D-Y were superior in children with health conditions than in healthy children.²⁴ Although following the EQ-5D-Y-5L translation protocol (and there being no reason to suppose that hierarchical ordering of severity levels would differ between sick and healthy children), it is recommended that future validation studies include more children with health conditions.

The strengths of this study include, firstly, being the first (to our knowledge) to pilot cross-cultural adaptation of the beta EQ-5D-Y-5L using the card ranking exercise as part of the draft EQ-5D-Y-5L translation protocol. The importance of rigorous translation and cross-cultural adaptation of HRQoL instruments cannot be overemphasized.²⁵ The use of a poorly translated instrument could invalidate otherwise carefully planned research. The card ranking exercise was instrumental in identifying and resolving severity level inversion through a series of interviews. Second, it is expected that older children/adolescents will have fewer problems in understanding the questionnaire. For this reason, the age of the children we interviewed was deliberately skewed to younger children (8–10 years) to establish comprehensibility among this younger age group. The next step in understanding the newly translated Chichewa EQ-5D-Y-5L version will be to test its psychometric properties in healthy, acutely ill, and chronically ill children.

Conclusions

A final adapted version of the beta Chichewa EQ-5D-Y-5L was somehow comprehensible to children in Malawi in terms of instructions and concepts covered by the measure. We recommend, however, that future users of the Chichewa EQ-5D-Y-5L should consider opting for the interviewer-assisted version rather than the self-complete version in children aged 8 to 10 years. Regarding concepts, the dimensions of “mobility,” “doing usual activities,” and “feeling worried, sad, or unhappy” clearly established conceptual equivalence and were understood by participants. There were, however, mixed results with comprehension of the “looking after myself” and “having pain or discomfort” dimensions. The EuroQol Group could consider including specific examples, particularly for the “looking after myself” dimension, to make this concept easier to understand by children in resource-limited countries or settings. Owing to its usefulness in this study, we recommend the use of the ranking exercise when developing translated versions of the EQ-5D-Y-5L in different cultural settings.

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Author Affiliations: Public Health Group, Malawi-Liverpool-Wellcome Trust Clinical Research Programme, Blantyre, Malawi (Ngwira); Department of Clinical Sciences, Liverpool School of Tropical Medicine, Liverpool, England, UK (Ngwira, Niessen); The EuroQol Group, Version Management Committee, Rotterdam, The Netherlands (Jelsma, Derrett); Department of Health and Rehabilitation Sciences, University of Cape Town, Cape Town, South Africa (Jelsma); Institute of Global Health Innovation, Imperial College London, London, England, UK (Maheswaran); Data Department,

Malawi-Liverpool-Wellcome Trust Clinical Research Programme, Blantyre, Malawi (Kapakasa); Department of Preventive and Social Medicine, University of Otago, Dunedin, New Zealand (Derrett); Nuffield Department of Primary Care Health Sciences, University of Oxford, Oxford, England, UK (Petrout); Department of International Health, Johns Hopkins School of Public Health, Baltimore, MD, USA (Niessen); Department of Health Services Research and Policy, London School of Hygiene & Tropical Medicine, London, England, UK (Smith).

Correspondence: Lucky G. Ngwira, MPH, Department of Clinical Sciences, Liverpool School of Tropical Medicine & Malawi-Liverpool-Wellcome Trust Clinical Research Programme, Chipatala Avenue, P.O. Box 30096, Chichiri, Blantyre 3, Malawi. Email: lgngwira@yahoo.com

Author Contributions: *Concept and design:* Ngwira, Jelsma, Maheswaran, Petrout, Niessen

Acquisition of data: Ngwira, Kapakasa

Analysis and interpretation of data: Ngwira, Jelsma, Kapakasa, Derrett, Smith

Drafting of the manuscript: Ngwira, Jelsma, Derrett, Petrout, Niessen, Smith

Critical revision of the paper for important intellectual content: Jelsma, Maheswaran, Kapakasa, Derrett, Petrout, Niessen, Smith

Statistical Analysis: Ngwira

Provision of study material: Derrett

Obtaining funding: Ngwira, Jelsma

Administrative, technical, or logistic support: Niessen

Supervision: Maheswaran, Petrout, Niessen, Smith

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